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Aero-Thermal Design and Analysis of Gas Turbine Combustion Systems: Current Status and Future Direction

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GE Aircraft Engines (GEAE) has introduced several new aeropropulsion gas turbine engines with improved combustion systems since 1994 including the CFM56 dual annular combustor (DAC), the CF6-80 low-emissions single annular combustor (LEC), the GE90 DACII, in addition to the dry low emissions (DLE) for the LM1600, LM2500 and the LM6000. GEAE has also made significant progress in the development and application of the semi-analytical mechanistic (SAM) models and anchored CCD during the combustion design and development process as described in this paper. GEAE is working with NASA in the development and validation of advanced turbulence and radiation models as part of the National Combustor Code (NCC). These activities have benefited the GEAE combustion design process which now uses routinely both the SAM and CCD-based methods unlike the most conventional "cut-and-try" design practice.

I Introduction

GE Aircraft Engines (GEAE) has recently introduced several new turbo-propulsions combustion products including the Low Emissions Combustor (LEC) for the CF6-80C, unstaged (rich-cup) Dual Annular Combustor (DAC) for the GE90 by further improving the LEC Single Annular Combustion (SAC) technology, and staged lean DAC for the CFM56-5B and -7B. Over the last several years, lower emissions have become one of the key characteristics of gas turbine engines for power generation and other industrial needs. A 'lean premixed' approach has been used in the development of the Dry Low Emissions (DLE) technology used in the LM1600 (aero-derivative engines used for Land and Marines applications), the LM2500 and the LM6000 engines. The LM6000 and the LM2500 combustors employ a triple dome design with staging of fuel and air flows to achieve lean-premixed operation from light-off to full power. Because of the higher bleed capabilities of the LM1600 engine, we are able to meet all the design requirements of this engine model with a two-dome DLE combustion system. The DLE engine produces low level of NOx/ CO/UHC emissions over a wide load setting. NOx emissions goals of 25 PPM have been successfully met at site rating conditions for the entire family of LM DLE products. Summary of these low emissions combustors is given in this paper.

Empirical/analytical combustor design approach has become GEAE's standard process for advance technology, new product introduction and current product support activities. Significant progress has been made recently in developing and validating advanced design process and tools as described in this paper, along with recommendations for future modeling activities.

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II Description of Modern Combustors

Figure 1 gives a cross-section of the CFM56 single annular combustion system used in several CFM engine models which have a wide range of sea-level, standard-day, static take-off thrust ratings between 82 and 151 kN, overall engine pressure ratio (OPR) of 21.3 through 31.2, and the fan by-pass ratio ranging between 5.1 and 6.8.

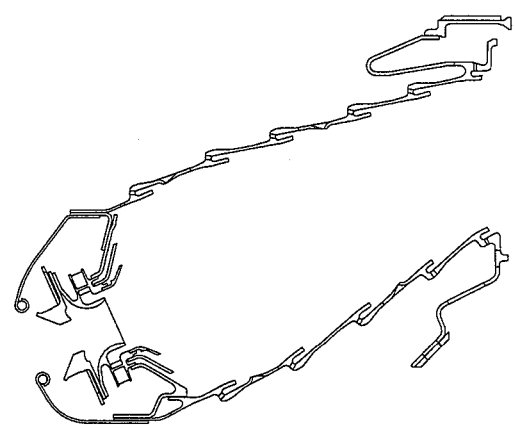


Figure 1 CFM56-2/3 combustion system cross-section.

The CFM56 single annular combustion (SAC) rich swirlcup technology developed since the early 1980's has been used in the various engine models - the latest model, the CFM56-7B

emissions was certified in July 1996. The sea-level static rated thrust specific fuel consumption (SFC) of these engine models vary between 9.1 and 11.03 g/kN-s. It is a common practice to plot the ICAO Landing-TakeOff (LTO) NOx emissions[1] as a function of the rated standard-day, sea-level, static takeoff pressure ratio (as shown in Figure 2) and assume that a linear relationship can best-fit the data. Consequently, the LTO NOx slope of the certified propulsion engines can vary significantly even for the same emissions technology level. For example, the calculated slopes for the CFM SAC combustors range between 1.6 and 2.8. However, by making simple assumptions, one can show that the slope of all these engine combustors is close to 2.2. This may be compared with the calculated slope of the V2500's LTO NOx curve of approximately 1.96. For a nominal takeoff pressure ratio of 30, the same combustion technology gives approximately 9% lower LTO NOx in the CFM56-5BP engine model compared to the CFM56-7B; the corresponding SFC values are 10.7 and 9.7 g/kN-s, also a 9% reduction approximately.

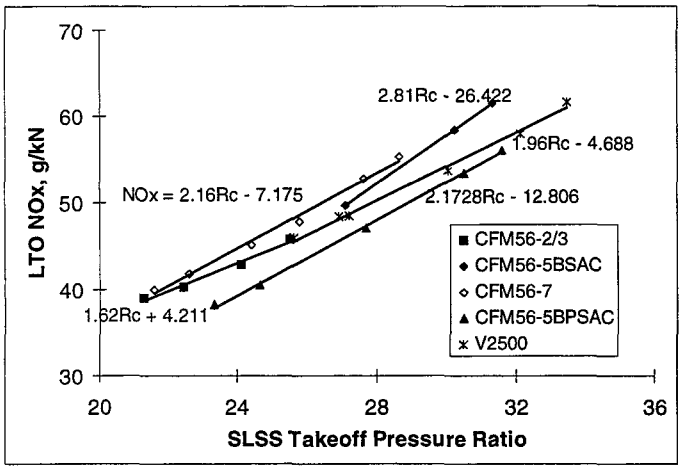


Figure 2 The Landing-Takeoff (LTO) NOx emissions levels of the CFM56 single annular combustors.

In order to further reduce NOx without adversely impacting other design requirements including low-power emissions, profile, operability, several staged combustion technology concepts were developed under NASA sponsored Experimental Clean Combustion Program (ECCP) and Energy Efficient Engine (E^3) technology programs [2-5]. One of the most promising approaches, namely Dual Annular Combustor (DAC), has been certified for the CFM56-5B (for producing low LTO and mission NOx emissions), and the high by-pass ratio (more than 8.0) GE90 (exceeding 40 OPR for some models) for designing a short-length combustion system compared to SAC with a single passage step diffusion system. Figure 3 shows a cross-section of the CFM56-5B DAC combustion system. As presented later in Figure 7, the CFM56 DAC combustion system produces the lowest levels of NOx emissions among the propulsion engines with the thrust ratings between 100-140 kN.

The CF6-45/50 family of engines (200-230 kN thrust class) was certified in 1979. A low NOx and HC combustion system as shown in Figure 4 was certified during 1985 for the CF6-80C, 230-270 kN thrust class family of engines. A low emissions combustion system (LEC) has been certified during 1995 that utilizes an improved swirl cup design, minimizes CO and HC quenching, and optimizes the primary and secondary zones to produce the lowest emissions single annular combustor for NOx, CO and HC and still retains design intent of the current CF6-80C combustion system including profile, performance and operability.

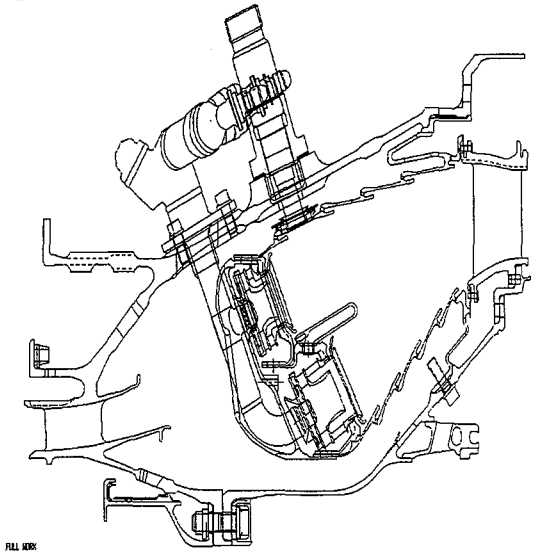


Figure 3 CFM56-5B Dual-Annular Combustion (DAC) system cross-section.

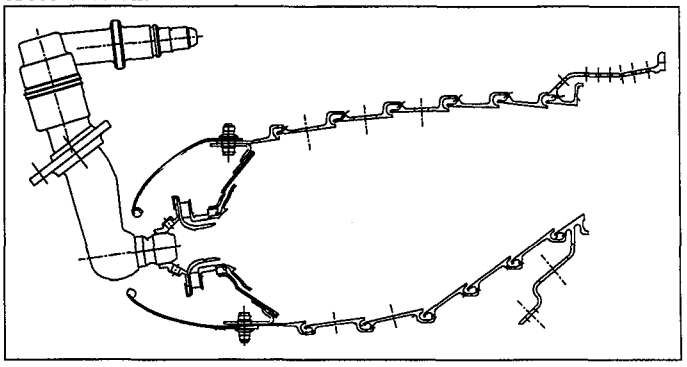


Figure 4 CF6-80C Single-Annular Combustion (SAC) system cross-section.

In order to reduce the combustion system length, the E^3 lean-combustion DAC technology for 30 OPR was applied for designing a near-stoichiometric combustion system (Figure 5) for the GE90, a 40 OPR engine for some of the thrust ratings. As expected, the GE90 DACI NOx emissions are comparable with that of the CF6 when OPR is extrapolated from 30 to 40, as shown in Figure 7. An improved combustion system, namely DACII has

subsequently been designed and certified to further reduce HC and CO significantly along with nominal reduction in NOx. This combustion system is simpler (requiring no fuel staging), and has further enhanced performance, profile and operability characteristics. The GE90 DACII is indeed the lowest total emissions engine among the nominal 40 OPR certified propulsion engines.

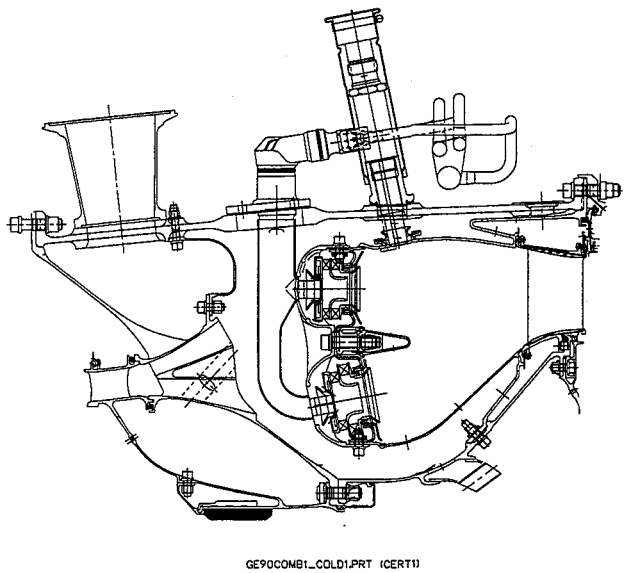


Figure 5 GE90 Dual-Annular Combustion (DAC) system cross-section.

Many of the industrial gas turbines for power generation, co-generation, and natural gas pumpers use water/steam injection to reduce NOx emissions at high power. Aero-derivative engines using dry-low emissions control technology, DLE have been installed recently at a number of customer installations. For example, the LM6000 DLE engine with a triple-annular combustion system, as shown in Fig. 6 (refer to Joshi et al.[6] for details on the LM6000 and other DLE models including the LM2500 and the LM1600) has been operating in the field since January 1995.

III Emissions of Turbopropulsion Combustors

There are several ways of comparing the gaseous emissions characteristics of propulsion engines. Mongia[7] summarizes the emissions levels of recently certified GEAE's engines. Figure 7 presents the landing-takeoff NOx emissions as calculated by the ICAO method and presented as a function SLSS takeoff engine overall pressure ratio. Compared to the single annular combustion technology of the late seventies (e.g., the CF6-45/50 at pressure ratio of 30), the modern engines (e.g., the

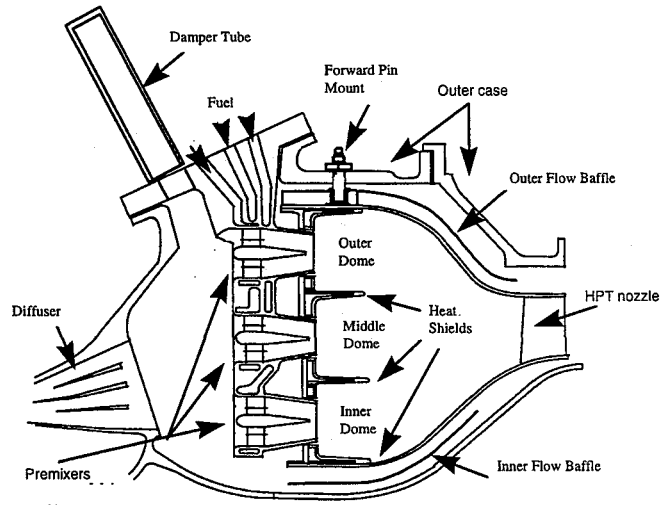


Figure 6 Cross-Section of the LM6000 Dry Low Emissions Combustor

emissions. Even though both the CFM56-5 and the CF6-80C2 use comparable single annular combustion technology, the LTO NOx emitted by the latter is 25% lower than that of the former. This is mostly due to the differences in the cycle conditions. Even the E^3 dual-annular combustion system with an advanced liner cooling technology (of "shingles") has higher NOx than the CF6; again due to the differences in the engine operating cycles.

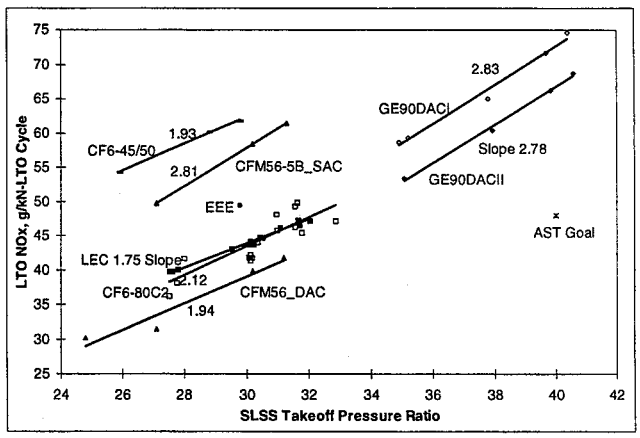


Figure 7 ICAO Landing-Takeoff NOx (LTO NOx) versus sea-level, standard-day, static (SLSS) rated takeoff pressure ratio.

CF6-80C2) emit 30% lower LTO NOx and the corresponding reduction in cruise NOx The CFM56-5B DAC combustion system gives 32% reduction in LTO NOx compared to the corresponding CFM56-5B SAC with comparable NOx-HC-CO technology characteristics, as discussed later in Figures 8 and 9, respectively.

Also shown in Figure 7 are the LTO NOx emissions of the GE90 DACI (staged combustor) and DACII that does not require any fuel staging between the inner and outer banks of the fuel nozzles. In fact, the GE90 DACII is an extension of the CF6-80C LEC SAC technology consistent with the objective of getting a shorter-length 40 OPR engine combustion system. Consequently, all the three engines, namely the CF6 pre-LEC, LEC and the DACII, fall essentially on the same line with a slope of 2.49. Historically, OPR has gone up approximately 5 every ten years. Therefore, one of the prime objectives of the NASA sponsored Advanced Subsonic Technology (AST) is to reduce LTO NOx of the future 50+ OPR engines to 50-30% of the current LTO NOx standards, namely 32 + 1.6*OPR. If one were to adopt successfully (with proper balance between cost, simplicity, performance, operability and emissions) the AST technology when scaled to 40 OPR, we will achieve 30 to 60% reduction in NOx.

The unburned and/or partially oxidized hydrocarbons emissions is strongly influenced by fuel nozzle spray quality (droplet size and velocity distribution, spatial and temporal concentration); fuel/air preparation; primary zone characteristics (ignition, flame stabilization and propagation; mean residence time, equivalence ratio and completion of combustion processes; the spatial and temporal variations in fuel/air ratio, temperature and velocity); post-flame and CO oxidation characteristics of intermediate combustion zone; interaction of the cooler film air (surrounding the dome and liner walls) with the core combustion zone. The combustion development experience during the last 20 years has shown us that there is a considerable level of tradeoffs required to meet low NOx/CO/HC/smoke emissions without adversely impacting other design requirements including durability, operability, combustor exit temperature quality, and pressure drop. In order to achieve the most optimum low emissions combustor, it has been a common practice to plot NOx versus HC emissions indices or preferably LTO NOx versus LTO HC.

Comparison of NOx-HC emissions technology of the three certified DAC's, namely the CFM56-5B DAC, the GE90 DACI and DACII is presented in Figure 8 along with that of the CFM56-5B Single Annular Combustor (SAC). It should be pointed out that whereas the CFM DAC is a lean primary-zone (PZ) combustion system (with attendant reduction in NOx emissions), the GE 90 DAC's have "near-stoichiometric" primary zone. In addition, the former have overall engine pressure ratio (OPR) close to 30, and the latter with 40 OPR.

During approach (30% rated thrust setting of the ICAO cycle points), the three high-thrust name plates of the CFM DAC fuel the both outer and inner banks (pilots and mains) of the nozzles, whereas only the outer bank is fueled for the lowest thrust name plate. The former and the CFM SAC fall along the same NOx-HC technology line and is slightly lower than that of the GE90 DACI. The lower thrust name plate DAC falls between the technology curves of the CFM SAC and the improved GE90 DACII. Approximately a 50% reduction in the CFM DAC HC emissions is possible if both the inner and outer banks of the fuel nozzles are

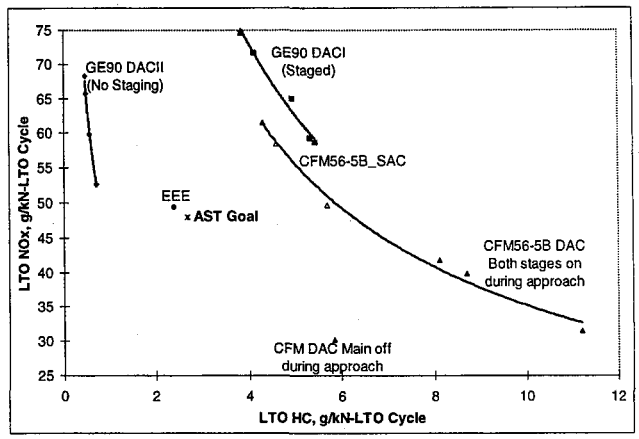


Figure 8 LTO NOx versus LTO HC of Dual Annular Combustors (DAC)

not fueled during approach. The E^3 technology (which used advanced liner cooling, namely shingles) is comparable with that of the CFM DAC when only the pilot stage is on during approach. However, compared to the CFM DAC with both stages fueled, the E^3 technology gives approximately 50% lower HC levels.

Comparison of the staged and unstaged "near-stoichiometric" GE90 DAC with the lean DAC's of the E^3 technology demonstrator and the CFM56-5B is presented in Figure 9. Also shown are the AST combustion technology goal, and LTO NOx and CO values of the CFM56-5B single annular combustor. A number of observations can be made here. First, NOx/CO technology of the CFM SAC and the lean DAC are comparable. Second, this technology currently falls between the current and improved "near-stoichiometric" GE90 DAC's.

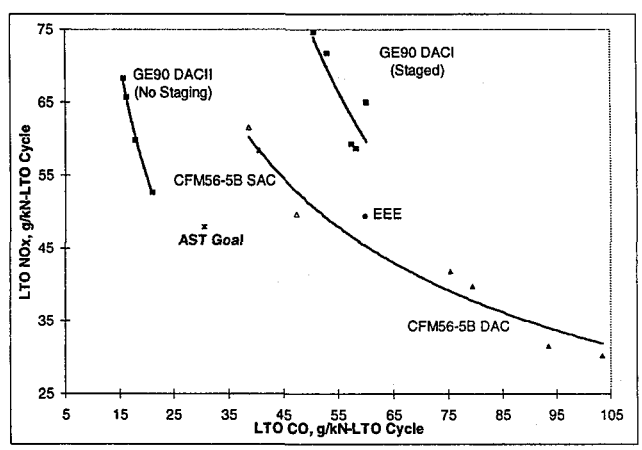


Figure 9 LTO NOx versus LTO CO of Dual Annular Combustors (DAC)

Third, the lean CFM DAC with both the main and pilot on (for the high-thrust name plates) and only the pilot fueled for the low thrust engine rating give the same NOx/CO characteristics. Fourth, the E^3 technology demonstrator gave slightly higher (14%) CO for a constant LTO NOx of 49.4. Fifth, the AST technology goal is to reduce CO by approximately 45% relative to the current lean DAC technology and product.

IV Emissions of Aero-Derivative Engines

A summary of emissions levels of the dry-low emissions aero-derivative engines recently introduced by GEAE (namely the LM1600, LM2500, LM2500+ and LM6000) is provided by Joshi et al.6. Table 1 lists power ratings, the simple cycle thermal efficiency, the engine air flow rates, pressure ratio, the nominal firing temperature and the emissions requirements of the LM1600, LM2500+ and LM6000 engines, respectively.

Table 1: Operational characteristics, and NOx / CO (ppmv corrected to 15% O2 dry in exhaust) emissions of the LM family of gas turbine engines (standard day, sea level, no inlet/exhaust losses.

	LM16000	LM2500+	LM6000
Power MW	15	29	44
Thermal Efficiency	37%	39.1%	42%
Air flow KG/Sec	43	77	125
Pressure ratio	22	22	30
Nominal Firing temp °C	1290	1290	1290
NOx/CO Emissions Requirements			
Synchronous idle		35/300	40/200
25% power		40/50	40/50
50% power	<50/25	<25/20	<25/20
100 % power	<25/20	<25/20	<25/20

The NOx emissions stay low as the combustor operate lean premixed over the entire operating envelop. CO emissions are within the 25 PPM guarantee levels at 50% power on up. UHC emissions are typically less than 10 PPM everywhere except during start-up. Higher emissions are encountered transiently during staging when additional domes are either being lit or extinguished. The NOx and CO emissions are shown as a function of power in Figure 9 for the LM6000 DLE combustor.

V Evolution of Combustor Design Methods

The gas turbine combustor design methods can be broadly divided into 1) empirical, 2) semi-empirical, 3) semi-analytical, 4) approximate or simplified models that capture "all" important and critical combustion processes (so called Semi-Analytical Mechanistic, SAM), and 5) analytical methods that make use of multi-dimensional Computational Combustion Dynamics numerical techniques (CCD) to design, analyze and refine combustion system. Empirical correlations

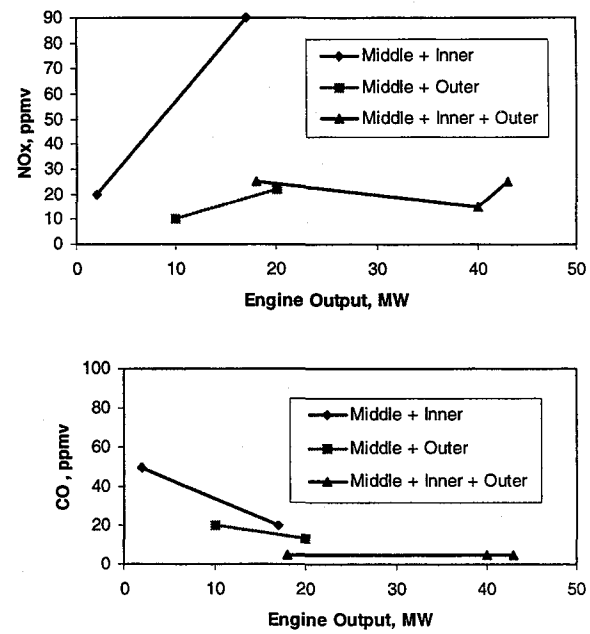


Figure 9: The LM6000 DLE Emissions Signature (ppmv, corrected to 15% O2 dry in exhaust)

for combustion efficiency, lean blowout (LBO), ignition, altitude relight, gaseous emissions have been used extensively by industry to conduct conceptual, preliminary, detail design and development of gas turbine combustion systems. Several empirical correlations have been published in the open literature; see Danis et al.[8] for the following empirical NOx emissions correlations as typical example:

$$NOx = 0.1972\, P_3^{0.4} \exp(T_3/194.4)$$
$$NOx = 0.0233\, T_4 P_3^{0.5} \exp(T_3/288) / V_{ref}$$

$$NOx = f\,[(P_3)^{0.5} \exp(T_3/345)]$$

A number of semi-empirical correlations, (e.g., References 8-11) have been proposed including the following for NOx:

$$NOxEI = 4.59 \times 10^{-9}\, F\, P_3^{0.25}\, \tau \exp(0.01 T_{st})$$

$$NOxEI = 1.5\times10^{15} (\tau - 0.5\tau_{evap})^{0.5} \exp(-71100/T_{st}) / [P_3^{\,0.05} (\Delta P_3/P_3)^{0.5}]$$

Several semi-analytical approaches have evolved over the last 25 years for correlating and predicting gaseous emissions. Mellor[10] gives an outstanding review of the early work in this area. In a more recent work, Rizk and Mongia[12] have used a reactor net-work(RNS) of perfectly stirred reactors (PSR), for correlating gaseous emissions from several gas turbine

combustors. One of the biggest disadvantages of RNS approach is that the modeled reactor volumes for the combustion zones of interest (e.g., primary, secondary and dilution zones) are significantly smaller than the corresponding physical volumes due primarily to the fact that these models essentially capture only the kinetics effects under the most ideal conditions.
Pratt[13], recognizing the deficiencies of the RNS modeling, proposed a gas turbine combustion model that took into account several critical processes and its refinements continued especially for jet-stirred reactors (e.g., Tonouchi and Pratt[14]). Tonouchi, Held and Mongia[15] have extended this approach to low-emissions rich-lean combustors. This SAM model takes into consideration micro-macro mixing, inlet non-uniformities and uses a more recently developed skeletal kinetic scheme for heptane by Held, Marchese and Dryer[16]. Held, Mongia and Tonouchi[17] have recently extended SAM to essentially premix flames of dry low emissions combustors of Joshi et al.[6].

A typical target for the accuracy goals for the empirical correlations has been close to that of the engine and/or rig data scatter, albeit the range of applicability of the correlation has been quite restrictive; this might even be reduced to the data interpolation. On the other hand, most of the published works on the semi-empirical and/or semi-analytical correlation have been limited to feasibility demonstration. Consequently, the accuracy and reliability of these correlations have not been good enough for providing significant advantage over the use of the most popular empirical correlations used for refining design modifications during combustion development process. Therefore, the designers have relied mostly on the use of empirical correlations for design, and on the semi-empirical/analytical correlations for getting the qualitative guidance.

The prime objective of the recent activities of Mongia and his coworkers (e.g., References 8, 15 and 17) is to improve the accuracy of SAM consistent with the target values listed in Table 2 so that SAM can be used effectively during the conceptual, preliminary and detail combustion system design phases, in addition to conducting a real-time test Data Reduction and Analysis (DRA), and reliable and timely response to field problems consistent with customers' expectations.

The first application of CCD coupled with empirical design approach was reported in References 18 and 19. This empirical/analytical design methodology has been used for designing 15 advanced technology combustors, four technology demonstrator engine combustors, three new centerline engine combustors, and one small turbine augmentor , in addition to providing valuable design insight in the following engine combustor design challenges, as summarized in References 20 and 21.

- wall carboning issues on the TPE331-10
- fuel spray and flow interaction in the ATF-3 combustor
- cold start and excessive hot wall temperature concerns on the recuperative engine GT601
- primary orifice placement optimization study on the TPE33-14/15 combustor
- development of a low-smoke TFE731 combustor
- root cause analysis for the 501-K34 convection/film cooled combustor liner distress
- low-smoke 570 combustor design
- T56-A427, T406/GMA2100 and GMA3007 combustor design changes

CCD has provided the qualitative design guidance similar to the semi-empirical/analytical correlations (as mentioned above) during the design process. Several attempts have been made during the last 20 years to improve accuracy of the CCD in regard to predicting performance, emissions, and liner wall temperatures; e.g., References 22-28. In addition, improved models (including semi-empirical, semi-analytical and comprehensive models) have been developed for fuel nozzles and the combustion diffusion system; References 29-34.

In order to move further, and SAM and CCD to provide quantitative guidance during design and development phases (with attendant benefit of minimal component and engine testing), References 35 and 36 have defined the accuracy target goals (e.g., Table 2) for future SAM and CCD modeling activities. The following sections give a status of the most recent activities being conducted by Mongia and his coworkers.

Table 2: Accuracy goals for next-generation gas turbine combustion models - approximate and CCD-based.

Variable	Predictive accuracy band
NOx	± 5 %
CO and unburned hydrocarbons	± 10 %
Exhaust SAE Smoke Number	± 2
SLS idle lean blowout fuel/air ratio	± 0.001
Pattern factor (Peak Profile Factor)	± 0.03
Average radial profile factor	± 0.015
Maximum liner wall temperature	± 25^0 F
Combustion system pressure drop	± 0.25 (% Pt3)
Diffusion system pressure drop	± 0.125 (% Pt3)
Circumferential location of the liner hot spot	± 0.1 × Nozzle sector
Axial location of the liner hot spot	± 0.01 × Comb. length

VI Status of Semi-Analytical Mechanistic Models

A summary of the semi-analytical mechanistic modeling activities of References 15 and 17 is given in the following paragraphs.

Tonouchi et al.[15] model a rich-dome gas turbine combustor by two reactors in series; the first is a partially-stirred reactor (PaSR) simulating the primary zone, and the second is a "partially-stirred" plug flow reactor (PFR) representing the dilution zone. The Gas Particle Monte Carlo (GPMC) simulation was initially based upon the micro/macro-mixing reactor model of Tonouchi and Pratt[14], and Tonouch [37]. Fluid flow and combustion are treated as the interaction of gaseous fluid particles which are "born" as they enter the combustor, "age" as they reside in the combustor, and "die" as they exit. The GPMC model accounts for many of the major processes in a gas turbine combustor, such as imperfect premixedness of the fuel and air inlet stream, finite-rate macro-mixing (the gross convective recirculation which brings the thermally hot fluid particles into proximity with the unignited cold fluid particles), and finite-rate micro-mixing (the molecular diffusion among particles that allows the hot particles to ignite the cold particles through the diffusion of heat and mass). The model also simulates other features in the combustor such as finite-rate mixing of dilution air jets and cooling film air.

The GPMC model uses a two-reactor approach to account for the fundamentally different flowfields within the combustor. The "primary zone reactor" is used to model the highly recirculating flow in the primary zone. The secondary reactor models the approximately plug flow of the gases as they exit the primary zone and pass through the intermediate and secondary zones. The potential of the GPMC model was demonstrated by calculating emissions from the three certified propulsion engine combustors.

Figure 10 presents the calculated and measured CO and NO_X emissions from one of the three combustors, identified as Combustor 1. The modeled results are the particle ensemble mean properties exiting the dilution flow reactor. The data in brackets next to the modeled results are the root mean square (rms) of the particle properties exiting the dilution flow reactor. The agreement with the measured data is reasonable, considering the assumptions and lack of optimization used in modeling these data. Better agreement with the measurements can be expected by considering the variation of the primary model parameters.

In a follow-on paper[17], the model is used to calculate NO_X and CO emissions from a natural gas-fired single cup dry-low emissions combustor of an aero-derivative industrial engine. This combustor provides high-quality benchmark data with few of the complications of a standard rich-dome propulsion

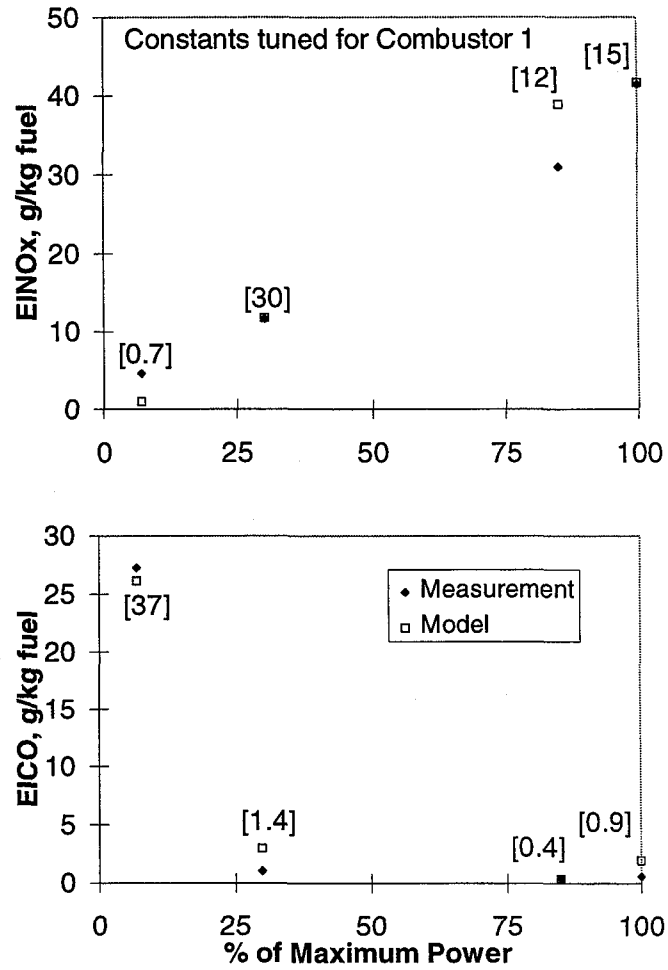


Figure 10. Comparison of the modeled results with the measured data for a rich cup propulsion engine Combustor 1.

engine gas turbine combustor (i.e., no liquid fuel/spray modeling, highly premixed, no dilution jets, etc.). A typical comparison between the GPMC and the semi-analytical model (represented by PSR and PFR) for both the NOx and CO shown in Figure 11 clearly demonstrates the advantage of SAM.

VII Status Of Anchored CCD

As stated earlier, the use of CCD for providing qualitative design guidance is wide-spread; the hybrid modeling conducted by Rizk and Mongia[23-33] has been a significant step of evolution for quantification of design requirements; Danis, Burrus and Mongia[36] are the first to publish comprehensive comparison between the measured and "anchored" predictions

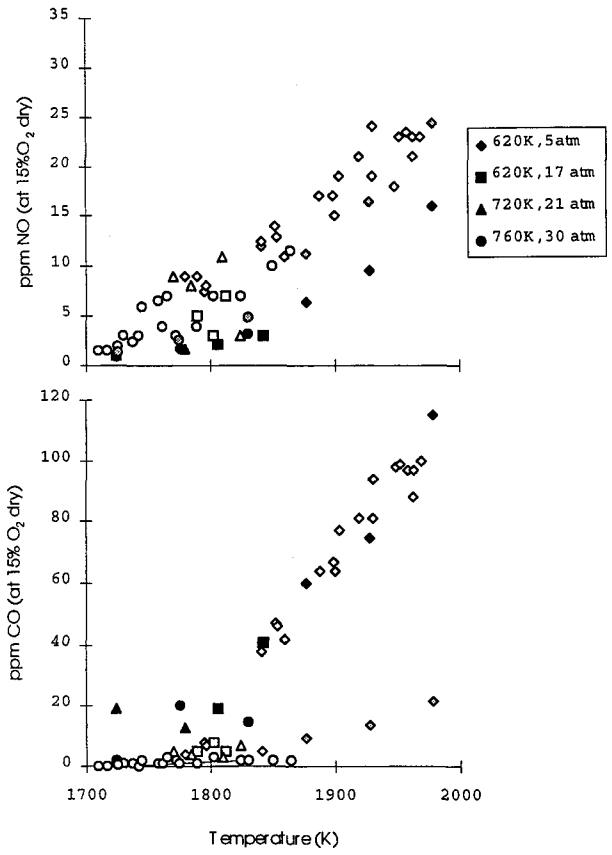


Figure 11. Comparison of GPMC model results with the measured data for the single-cup combustor. The open symbols are measured data, the solid symbols are GPMC calculations, and the gray symbols are PSR + PFR calculations

of combustor exit profile/pattern factor, lean blowout (LBO) fuel/air ratios, NOx, CO and unburned hydrocarbons (HC) of recently certified propulsion engines, and apply the "anchored CCD methodology" to the GE90 DACII combustion system. This approach gives, in addition to qualitative picture of the flowfield, (Figures 12 and 13 respectively for u-v velocity vectors and temperature field), the predicted distribution of NOx emissions (Figure 14), CO and HC profiles accurate enough to claim good agreement with LTO NOx values for several recently certified propulsion engines as summarized in Table 3.

The majority of the most recent emphasis has been on establishing how far one can go for the rich swirl-cup unstaged diffusion flame combustors (e.g., the GE90 DACII and the CF6-80C2 LEC with the state-of-the-art turbulent combustion models; namely, fast-chemistry approximation for high-power, 2-step eddy-breakup model for low-power[19, 38], standard

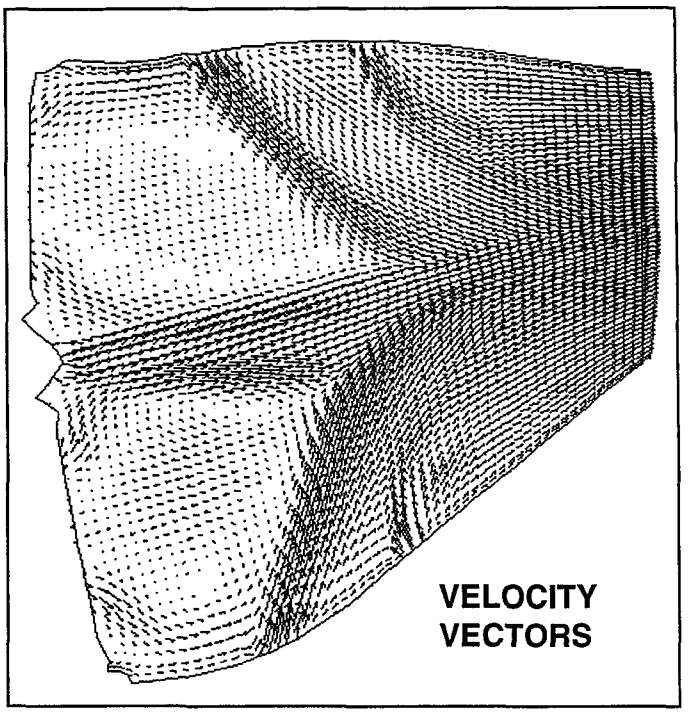


Figure 12. Predicted velocity field characteristics provide insight for optimizing combustor's conflicting design requirements

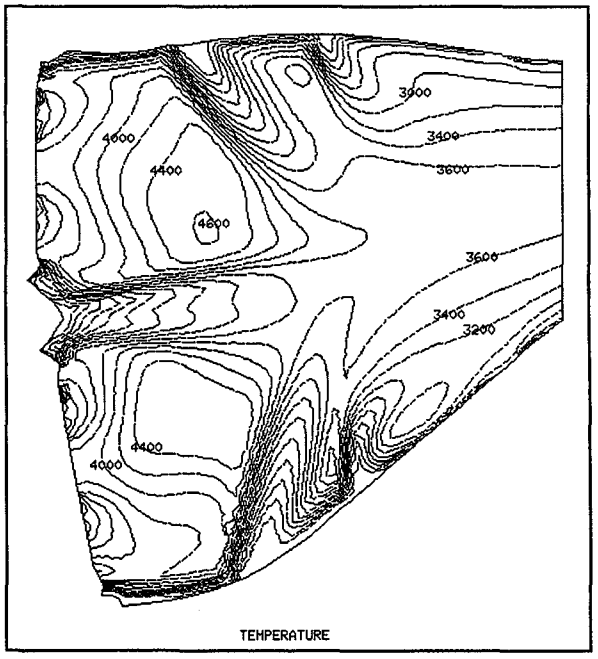


Figure 13. Predicted gas temperature distribution.

k-ε turbulence model, deterministic spray treatment etc. The difference between predicted and measured values

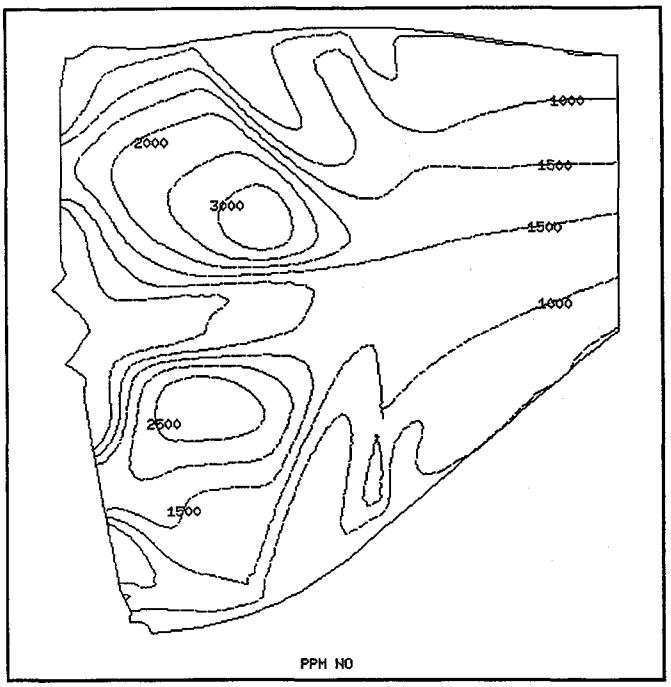


Figure 14. Predicted NO contours.

Table 3. Comparison between measured and rich swirlcup anchored predictions for the ICAO Landing-Takeoff emissions levels.

COMBUSTOR	AVERAGE ENGINE LTO:	ICAO DATA (g/KN)	ANCHORED PREDICTION (g/KN)
GE90-92B DAC-2	NOx	70.7	70.7
	CO	14.4	13.8
	HC	0.5	0.4
	SMOKE	0.5	-
GE90-92B DAC-1	NOx	73.1	67.2
	CO	55.5	29
	HC	4.4	9.9
	SMOKE	9.3	-
CF680C2 LEC	NOx	46.5	44.4
	CO	23	-
	HC	1.8	-
	SMOKE	6.9	-
CFM56-5B SAC	NOx	61.5	58.5
	CO	38.6	-
	HC	4.3	-
	SMOKE	10.1	-

of LTO NOx for the GE90 DACII, GE90 DACI, the CF6-80C LEC and the CFM56-5B SAC are 0, -8.1, -4.5 and -4.9%, respectively, considered within the range of experimental variability. Even though the emphasis has been for correlating CO and HC for the GE90 DACII (where the difference between calculated and measured LTO CO and HC are -0.6 and -0.1 g/kN, respectively), we expect a similar level of agreement for the CF6-80C LEC and the CFM56-5B SAC. One however needs to use improved turbulent combustion models for achieving acceptable levels of agreement for staged combustion, e.g., the GE90 DACI near-stoichiometric combustor and the lean-staged CFM DAC. We will discuss this later in this paper.

The anchored CCD approach of Danis, Burrus and Mongia[36] has achieved good agreement with, e.g., combustor exit temperature quality as shown typically in Figure 15 for the profile and pattern factor.

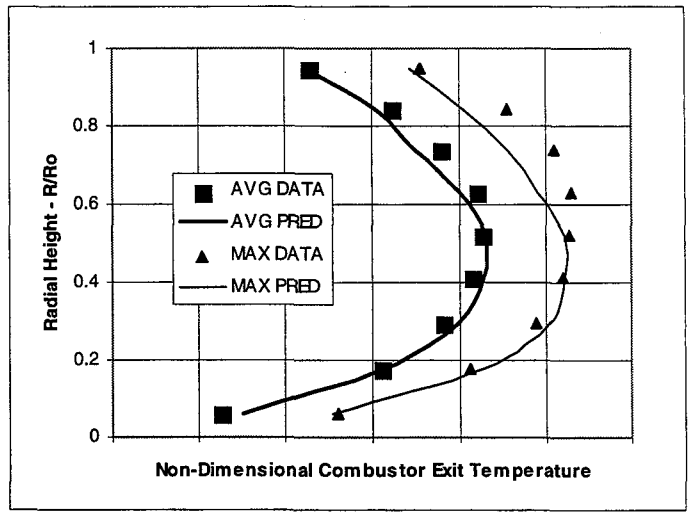
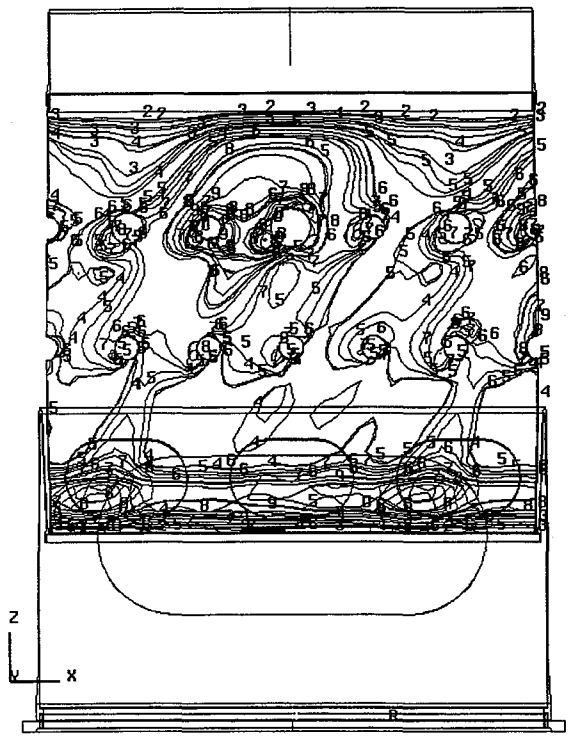


Figure 15. Typical comparison between predicted and measured average and maximum temperature profiles at the burner exit plane.

A state-of-the-art quasi-1-D liner and dome structure temperature technique (e.g., Kumar et al.[40]) has been extended to quasi-3-D structure calculations that use the combustor internal flow calculations, coupled with wall functions, measured film effectiveness for both the hot-side and cold side along with appropriate heat transfer coefficients, PATRAN and P-Thermal commercially available codes. Typical liner wall temperatures by this approach as shown typically in Figure 16 are similar to that of Rizk[39], except the latter did not simulate the structural details required for predicting stress levels as are done in this study. Clearly, the typical predicted liner temperature distribution show significant gradients; and hence the need for grid resolution via PATRAN or equivalent software package.

We have used the quasi-3-D structure temperature calculation for the recently certified GEAE engine combustors to establish its capability, as shown for example in Figure 17. Here we



Contour #	Degrees F	Contour #	Degrees F
8	1803	7	1729
6	1655	5	1581
4	1507	3	1433
2	1359	1	1285

Figure 16. Typical predicted liner wall temperature contours.

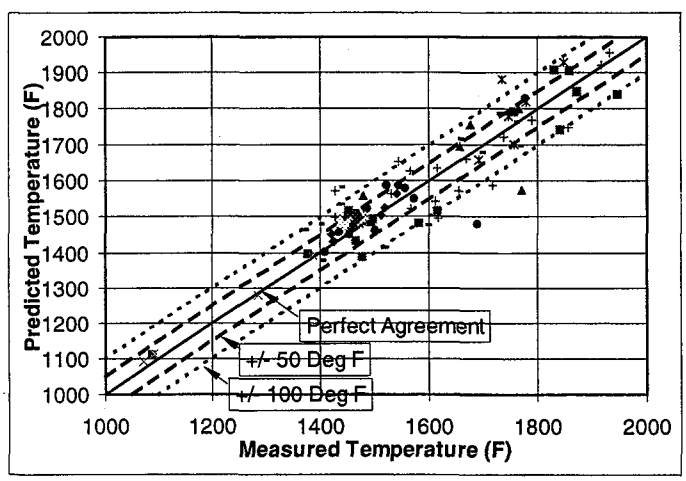


Figure 17 Comparison between predicted and measure wall temperatures with GEAE's anchored rich swirlcup CCD

present predicted wall temperatures versus measurement, clearly showing that the current technique can calculate the structure temperatures within an accuracy of ±100 F as compared to ±25 F goal (see Table 2).

VIII Future Direction

Having established Generation 1 SAM and anchored CCD methodology, we have recently initiated several activities to bench mark and/or improve:

a) CFD capability of the design code
b) Baseline turbulent combustion models based on eddy-breakup (EDC) and scalar transport PDF
c) Turbulence models relevant to gas turbine combustion
d) Radiation model
e) Spray model
f) Laminar flamelet modeling with reduced and skeletal mechanisms
g) Large eddy simulation (LES).

The following paragraphs give brief description of our activities in each of these eight areas.

An Advanced Combustion Code (ACC) is currently being developed to further advance the use of Computational Combustion Dynamics (CCD) at GE. ACC will have complex geometry representation capability via unstructured multiblock grids, and will comprise of a number of advanced combustion/spray/turbulence models that are being developed in a modularized fashion. The ACC solver is based on a fully implicit pressure correction algorithm with colocated grids. An overview of an early version of ACC is given by Tolpadi et al.[41]. This code will allow a more accurate representation of staged combustion systems including DLE as shown in Figures 18 and 19.

Since the first use of CFD in combustion design [18,19], we have felt strongly for the need to improving turbulence models for gas turbine combustion. In spite of concerted effort by several turbulence modeling groups over the last 25 years, our predictive accuracy for turbulent combustion field has not improved. Even after having expended a considerable effort on carefully measuring and modeling isolated subcomponent flowfields relevant to gas turbine combustors, we have not achieved any significant improvement in the turbulence modeling accuracy, e.g., see References 42 and 43. Even for the reacting flows where the turbulence model is expected to do a reasonably good job, we have not been to achieve accuracy target levels (Table 2) desired by the designers. This point is illustrated by the following test case.

A nonswirling jet diffusion flame in an axisymmetric combustor was experimentally investigated data for this flame by Lewis and Smoot[44]. Natural gas fuel was introduced centrally (0.016m. diameter port) and a coaxial diameter 0.0222m., outside diameter 0.0572m.) annular air stream was

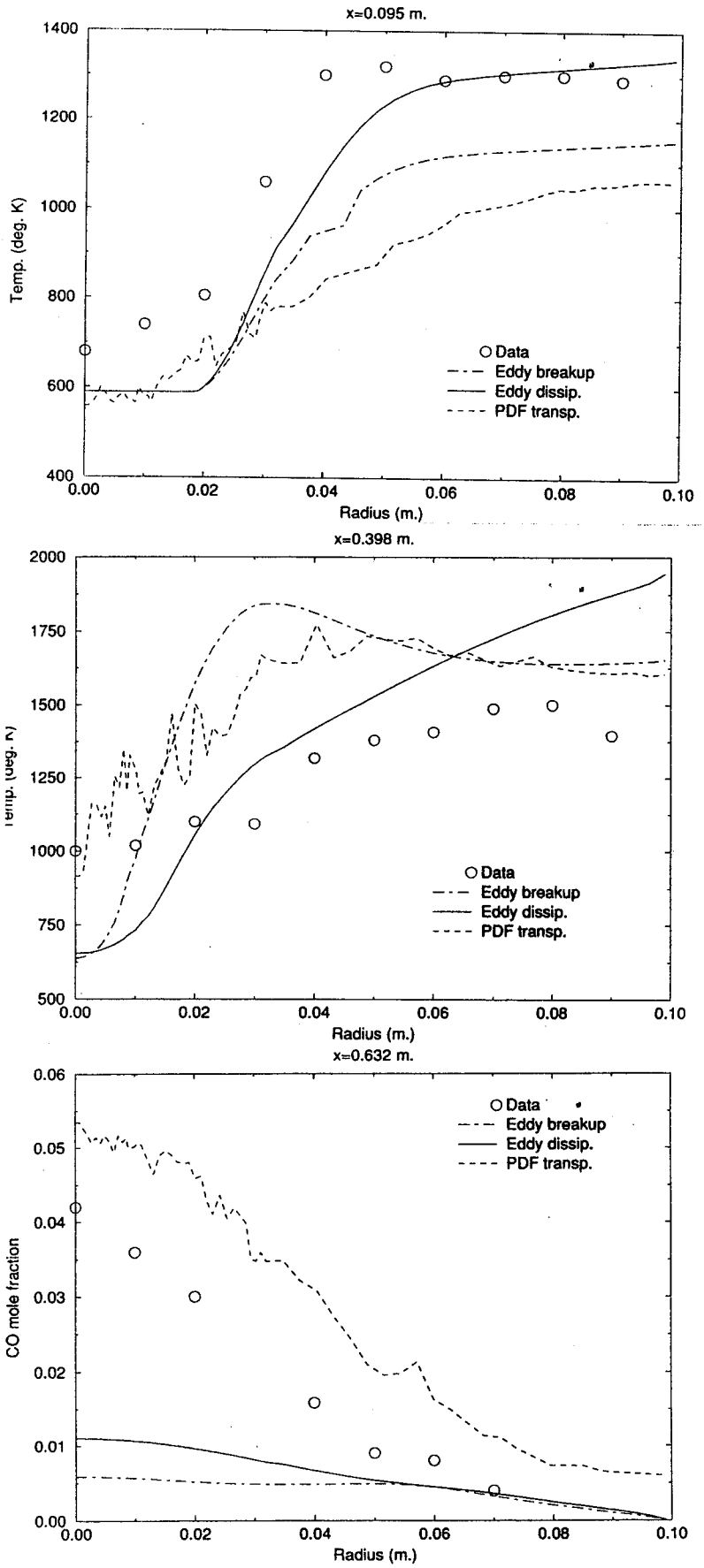


Figure 20 Typical comparison between data and predictions from eddy-breakup, EDC and scalar PDF transport models, from Tolpadi et al. (1998).

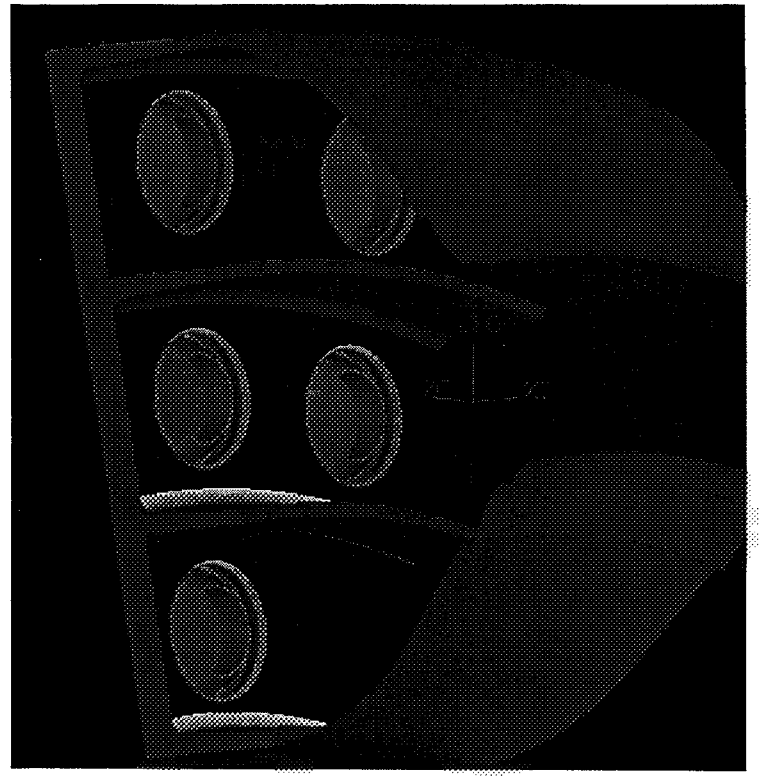


Figure 18 UG solid model of the LM6000 DLE combustor

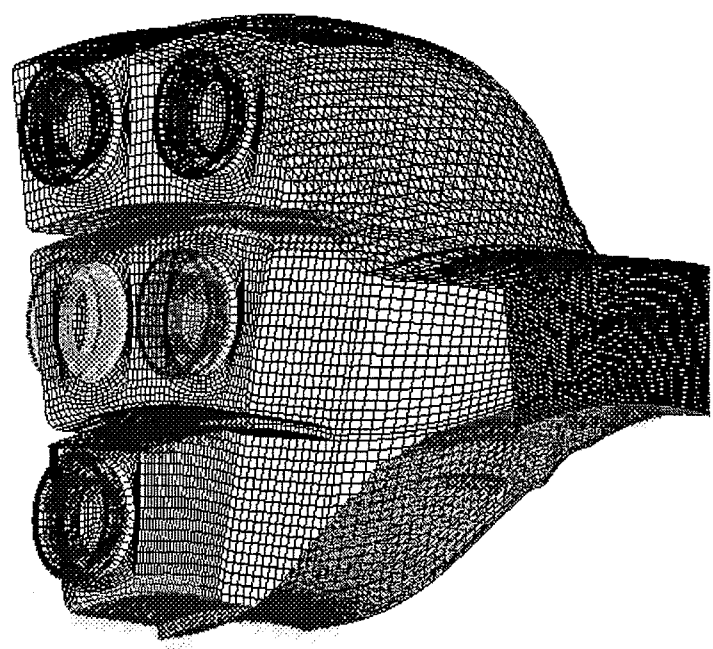


Figure 19 LM6000 DLE unstructured combustor grid of size 270,000 nodes

introduced into a vertical chamber of diameter 0.2032m. The incoming fuel velocity was 21.3 m/s with a temperature of 300 deg. K. The incoming air velocity was 34.3 m/s with a temperature of 589 deg. K. The pressure was atmospheric. Gas samples were taken at different axial and radial locations downstream and the specie concentrations were measured.
Numerical calculations of the jet diffusion flame were performed by Tolpadi et al[41] using the eddy breakup, eddy dissipation and the scalar PDF transport combustion models with the standard k-ε model. Even though one can offer several good reasons for the failure of the models to achieve acceptable level of quantitative agreement for such a simple test case , as shown for typical axial stations (95, 398 and 632 mm downstream, or the corresponding downstream distance of 1.7, 7.0 and 11.0 swirler diameters), it is nevertheless a discouraging state of the turbulent combustion modeling.

Encouraged by recent advances in turbulence modeling[45-50], we have recently started reassessing our strategy for improving turbulence modeling. For example, Nikjoo and Mongia[51,52] have compared the predicted profiles of axial velocity, turbulence kinetic energy and shear stress under the requirement that the predicted skin friction must agree with data for the fully developed pipe and channel flows in addition to the other standard test flows run to validate turbulence models. They also ran the turbulence models for a fully-developed pipe flow (Laufer (1954), Re_D = 50,000) with a specified flow as computed with the turbulence model of Yang and Shih[47] so as to establish the baseline comparison of these models for predicting shear velocity u_τ, centerline velocity U_c/u_τ, and pressure gradient dp/dx. As shown in Table 4, the values computed by the both methods are quite different for the popular turbulence models.

Table 4 Comparison of turbulence models for fully developed pipe flow, from Nikjoo and Mongia (1998)

Method 1: When turbulent Re number is matched

Turbulence model	Shear velocity u_τ	Centerline velocity U_c/u_τ	Pressure gradient dp/dx
L_S model	0.0389	20.935	-6.062E-3
Y_S model	0.0389	22.938	-6.06E-3
Chien model	0.0390	25.918	-6.07E-3
Wilcox	0.0390	19.225	-6.07E-3

Method 2: When mass flow rate is matched

Turbulence model	Shear velocity u_τ	Centerline velocity U_c/u_τ	Pressure gradient dp/dx
L_S model	0.0442	21.16	-7.81E-3
Y_S model	0.0396	23.35	-6.28E-3
Chien model			
Wilcox	0.0485	19.4	-9.43E-3

We are conducting a thorough assessment of these turbulence models against the benchmark quality database (e.g., the flow over a backward facing step, sudden pipe expansion, confined swirling flows, film effectiveness of multi-hole and slot film-cooling techniques) to achieve accuracy levels consistent with Table 2, References 51-53.

A non-gray sum-weighted, narrow-band radiation model consistent with finite-volume 3-D gas turbine combustion model has been formulated and coded for incorporation as a module to the National Combustion Code being developed jointly by NASA and industry. Kumar et al.[54] have done preliminary assessment of this model. In a follow-on paper (Kumar and Mongia[55]), comparison has been made between the narrow-band wide-band radiation models with the state-of-the-art wall temperature calculation method[56] that uses empirical correlation for emissivity along with the popular luminosity factor expressions. As shown in Table 5, the wall temperature levels calculated by the wide band model (which is computationally faster than the narrow-band model by a factor of more than 5) are quite close to those of the narrow-band model for a typical liner at 40 atm. and 0.03 fuel/air ratio; the resulting root-mean-square (RMS) error is quite small. On the other hand, the differences between the narrow-band and the conventional state-of-the-art models are substantial.

Table 4 Comparison between narrow-band (NB), wide-band (WB) and conventional state-of-the-art[56] radiation models, from Kumar and Mongia (1998)

Radiation models	Max Liner Temperature, F	Average Liner Temp., F	RMS Error, F
Narrow-Band	1922	1505	0
Wide-Band	1914	1502	4.6
Conventional	1977	1592	94.3

Simple empirical, semi-analytical and comprehensive CFD-based fuel nozzle models[29-33] need to be further developed in regard to the film breakup details and other process critical for improving fuel/air mixing and validation with data[56, 57]. The improved understanding gained as a result of recent experimental and analytical studies pertinent to droplet and spray turbulent convective transport, heat-up and evaporation modeling studies are being incorporated[58] in the Advanced Combustion Code[41].
Having established the baseline anchoring capabilities with the state-of-the-art CCD, e.g., Table 3, Figures 15 and 17, we have started looking into how far we can go with the anchoring methodology for the dry-low emissions (DLE) combustion systems in regard to predicting flow field[59, 60], NOx and CO[60, 61], dynamics[62], consistent with the database[6, 63]. The feasibility

of extending a recently formulated dynamics model[62] for DLE to a staged diffusion-flame combustor has been reported in Reference 64. The following paragraphs give a brief description of these modeling activities.

Laser Doppler Velocity measurements of reacting and non-reacting flows in a rectangular duct equipped with a single LM6000 or LM2500 premixer were conducted at GE CR&D. The experimental set up consisted of a single production premixer cup installed in a high pressure vessel built specifically for application of advanced diagnostic techniques to single cup studies (see Figure 21) and allow comparison with model predictions.

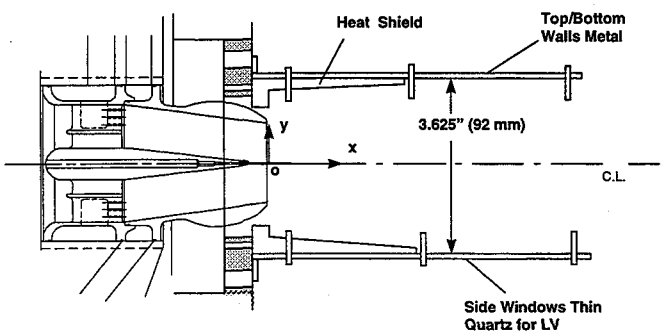
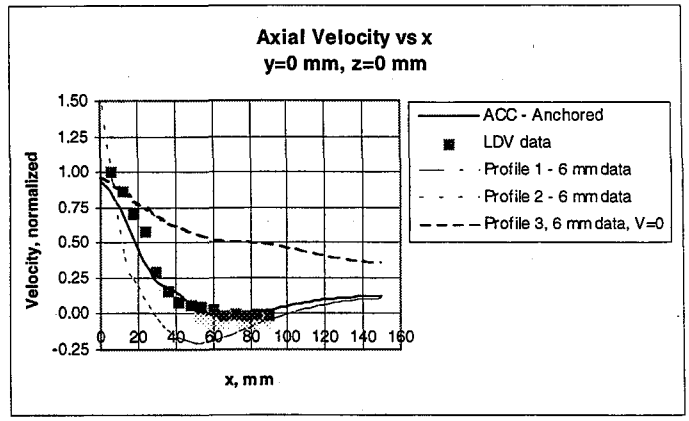


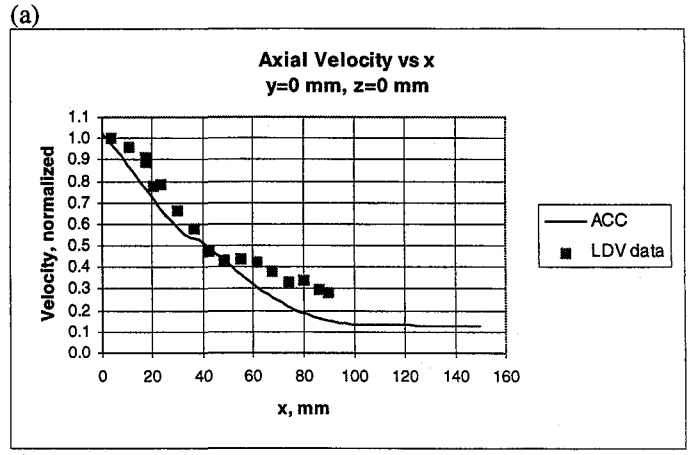
Figure 21 Experimental set-up for LV measurements of the LM6000 single cup premixer (from Hura et al. (1998)).

Hura et al.[59] were able to anchor their two-step eddy breakup model after spending considerable effort to achieve reasonable agreement (denoted as ACC Anchored in Figure 22) with the axial velocity component along the mixer's center-line. A laminar flamelet model has been used by Held and Mongia[60] to demonstrate its feasibility for predicting the flow-field and NOx emissions levels of the LM6000 DLE mixer, as shown in Figures 23 and 24; followed by CO prediction and comparison with data in Reference 61.

Under partial sponsorship of the AFOSR Focused Research Initiative, Menon and his associates have been working on the use of large eddy simulation with linear eddy (LESLE) subgrid modeling[65-68]. The feasibility of his approach for the LM6000 DLE mixer (that has been modeled by Hura et el.[59] and Held and Mongia[60, 61] by the 2-step eddy-breakup, EBU and laminar flamelet modeling, respectively, as shown in Figures 22-24) has recently been demonstrated[69]. He used the same inlet profiles as anchored by Hura et al[59]. His predicted centerline axial velocity (Figure 25) should be compared with the EBU results of Hura et al. Shown in Figure 22a. This is a significant step forward with LESLE, considering that these calculations have been made by Menon with no prior knowledge of the



(a)



(b)

Figure 22 Comparison between measured and "anchored" centerline velocity of (a) the LM6000 DLE mixer, and (b) the LM2500 DLE mixer with the eddy breakup model (from Hura et al. (1998)).

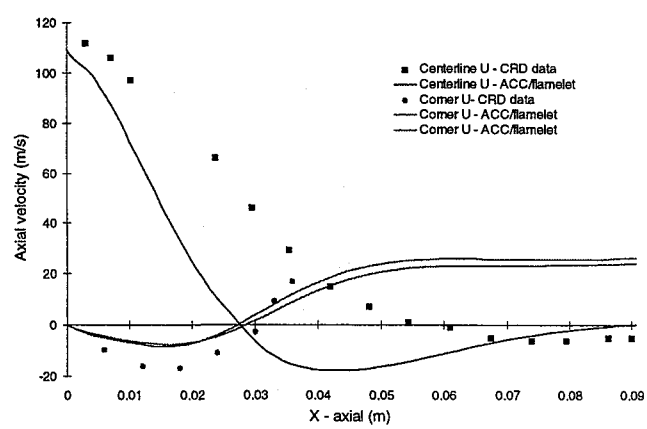


Figure 23 Comparison between measured and "anchored" centerline and corner axial velocity component of the LM6000 DLE mixer with a laminar flamelet model (from Held and Mongia (1998)).

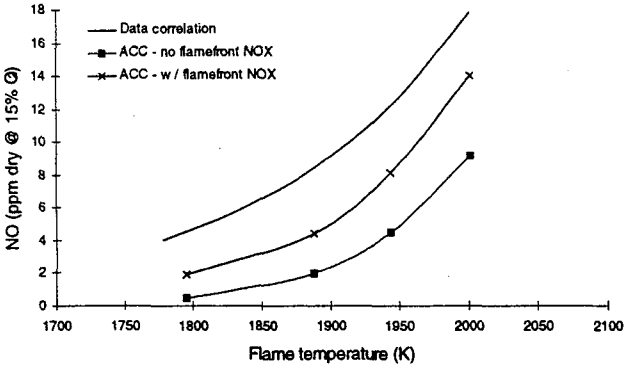


Figure 24 Feasibility demonstration of NOx prediction with a laminar flamelet model and comparison with data of the LM6000 DLE mixer (from Held and Mongia (1998)).

experimental values, and no attempt has been made to "further tune" LESLE's model predictions with the data.

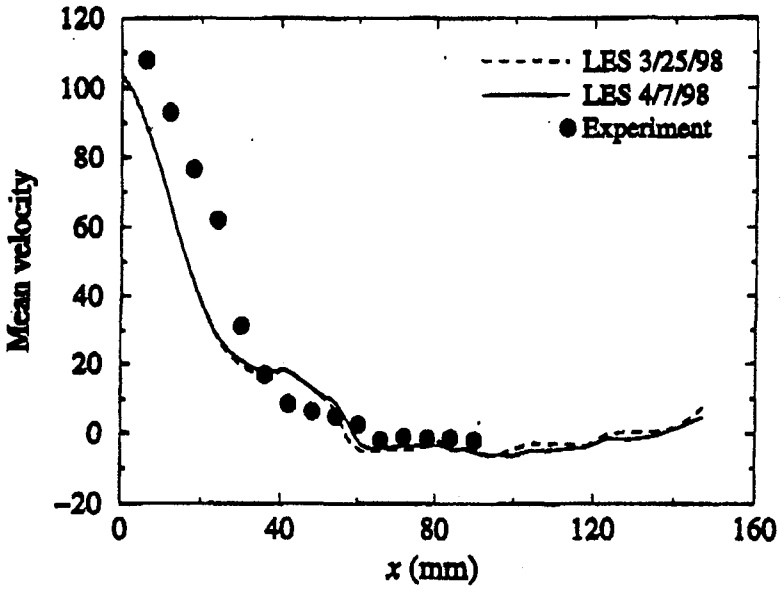


Figure 25 Feasibility demonstration of Large Eddy Simulation (LES) for predicting the LM6000 DLE mixer flow field - Centerline axial velocity (from Menon[69]).

One of the biggest advantages of the LESLE's calculations is that it provides dynamic structure of the flame, as shown in Figures 26 through 28, where time-averaged vorticity plots are compared with the five randomly selected instantaneous time snap shots. This type of information along with measured data, for example from Peters[70], another AFOSR FRI sponsored work for a generic DLE mixer, shown in Figure 29, give the research (leading to design) tools for providing the insight required to develop technology in a more cost-effective timely manner.

IX. Summary

Several low-emission technology combustors, including the CFM56-5/7B dual annular combustors, the CF6-80C2 low-emissions single annular combustor, and the GE90 unstaged dual annular combustor (for turbopropulsion engine applications), the dry-low emissions staged multi-dome annular combustors for aero-derivative industrial engines (the LM1600, LM2500 and LM6000) have recently been introduced by GEAE. These products have significantly reduced the gaseous emissions as summarized in this paper. In addition, significant progress has been made in the development of semi-analytical mechanistic models and the CCD-based anchoring technique to help combustion designer with their tasks of designing and servicing the products to meet customers' needs.

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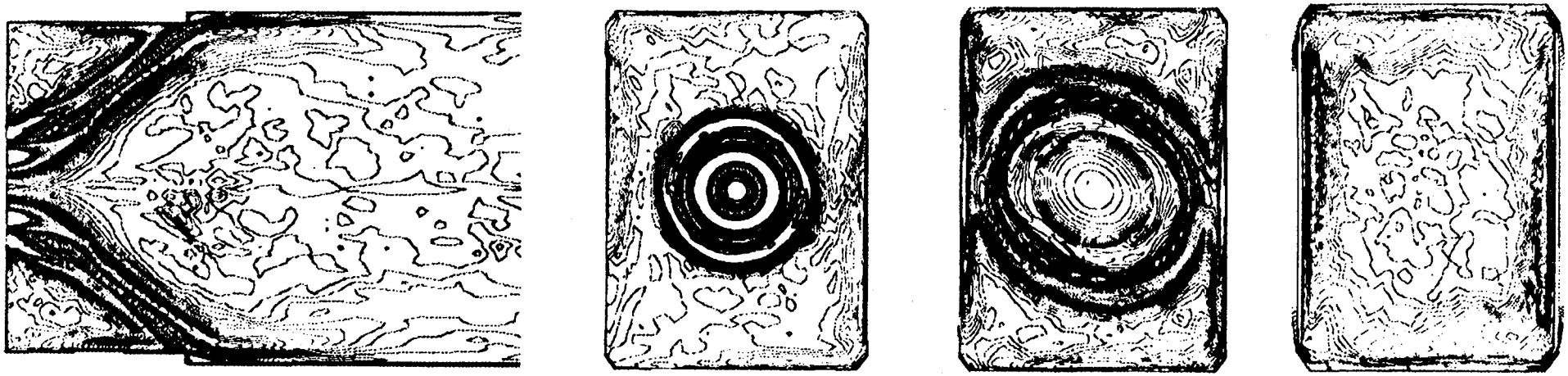
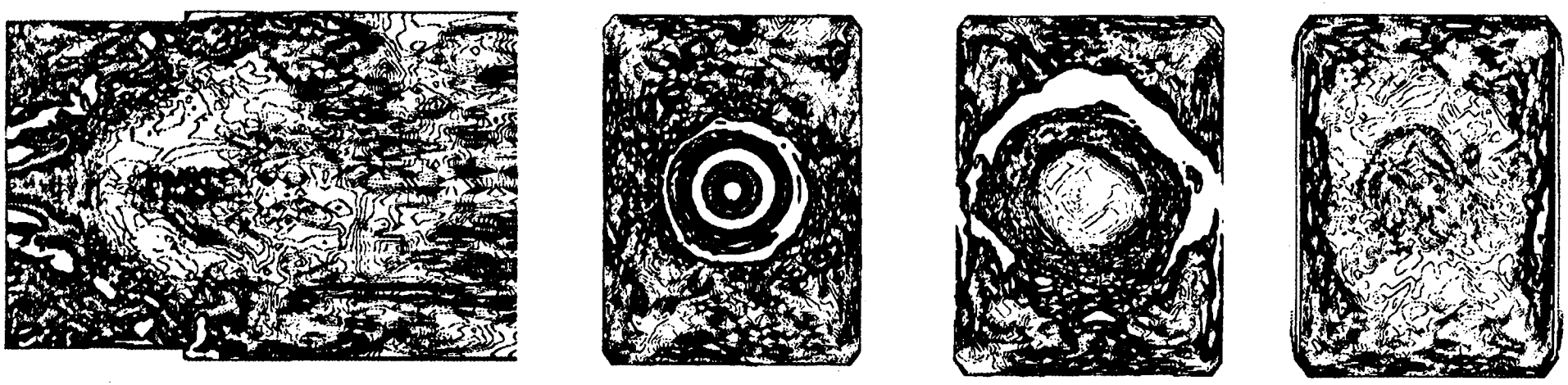
Time Averaged Vorticity Field



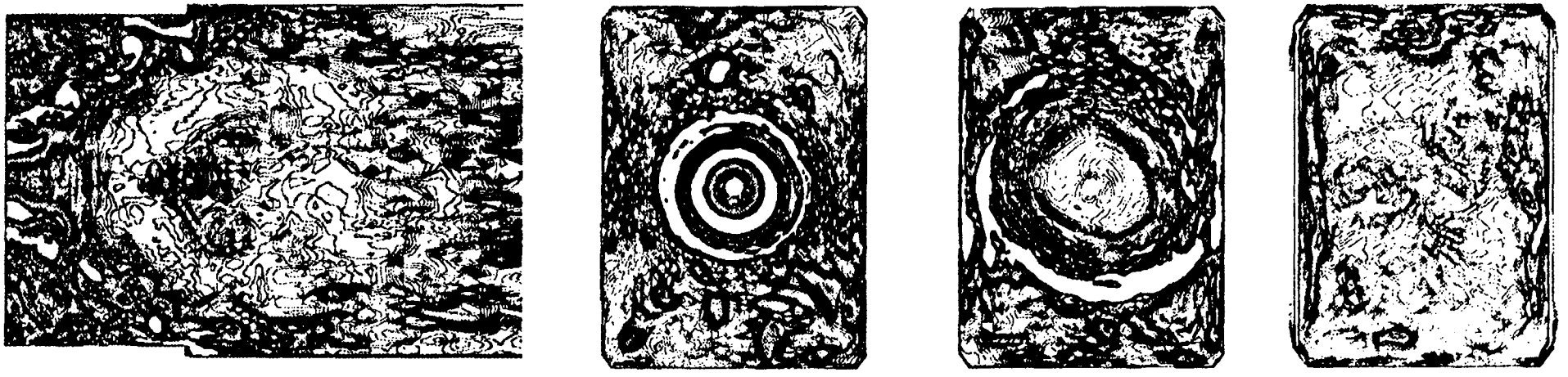
Instantaneous Vorticity Field #1



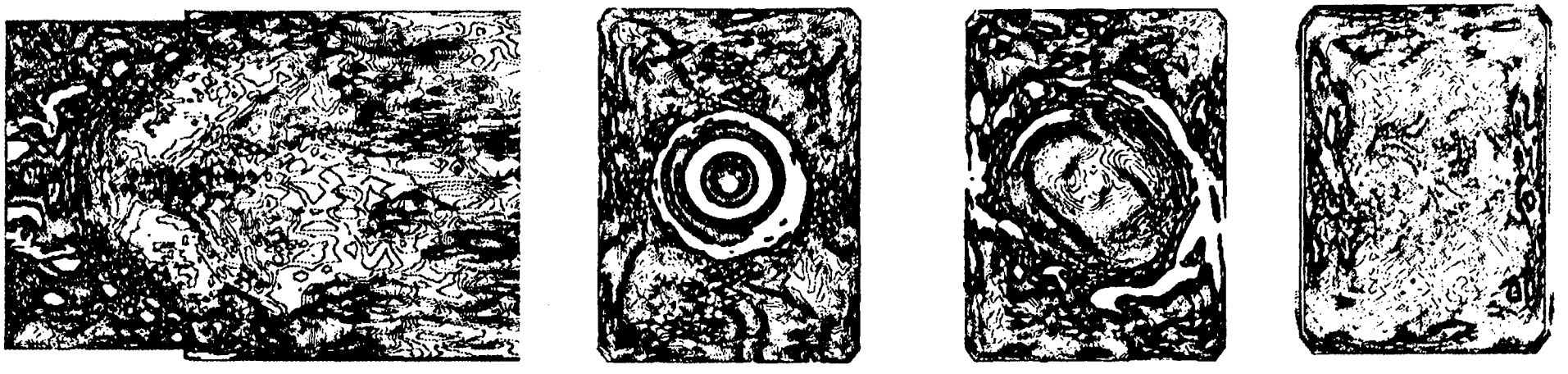
Side View **End View (X=6mm)** **End View (X=24mm)** **End View (X=78mm)**

Figure 26 Time averaged and instantaneous LESLE computed vorticity field for the LM6000 DLE mixer computed by Menon[69].

Instantaneous Vorticity Field #2



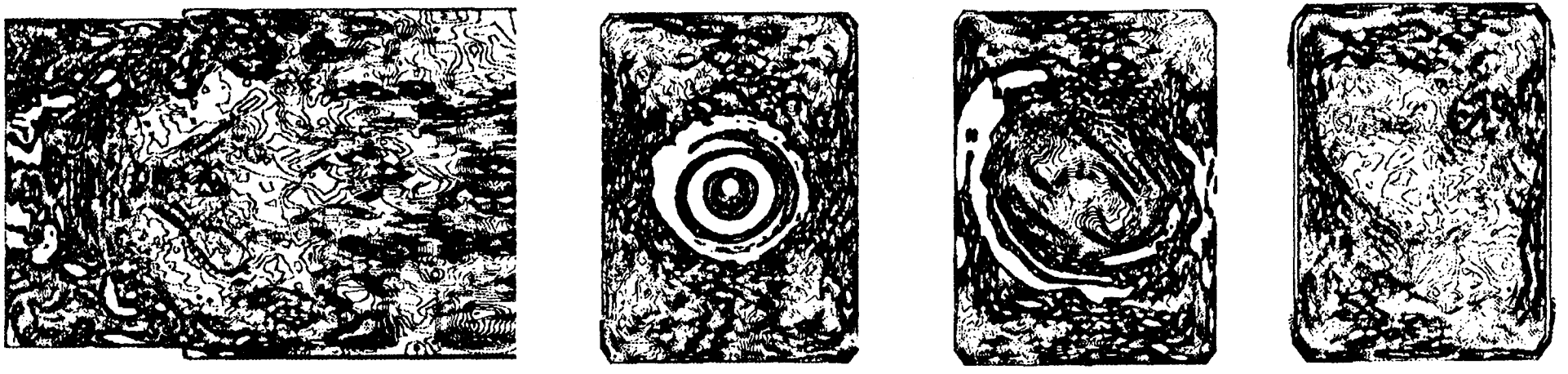
Instantaneous Vorticity Field #3



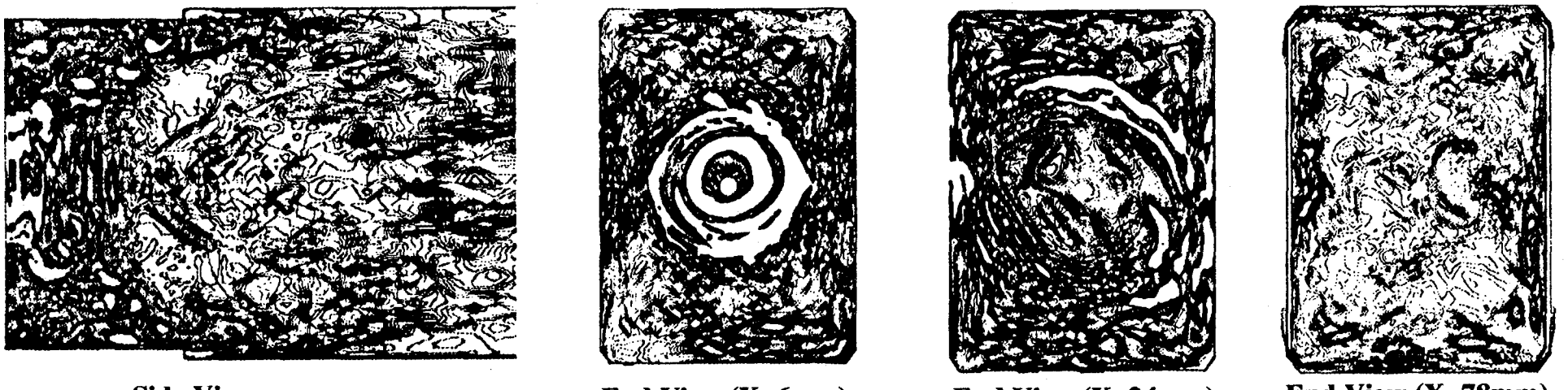
Side View **End View (X=6mm)** **End View (X=24mm)** **End View (X=78mm)**

Figure 27 Instantaneous LESLE computed vorticity field for the LM6000 DLE mixer computed by Menon[69]

Instantaneous Vorticity Field #4



Instantaneous Vorticity Field #5



Side View **End View (X=6mm)** **End View (X=24mm)** **End View (X=78mm)**

Figure 28 Instantaneous LESLE computed vorticity field for the LM6000 DLE mixer computed by Menon[69]

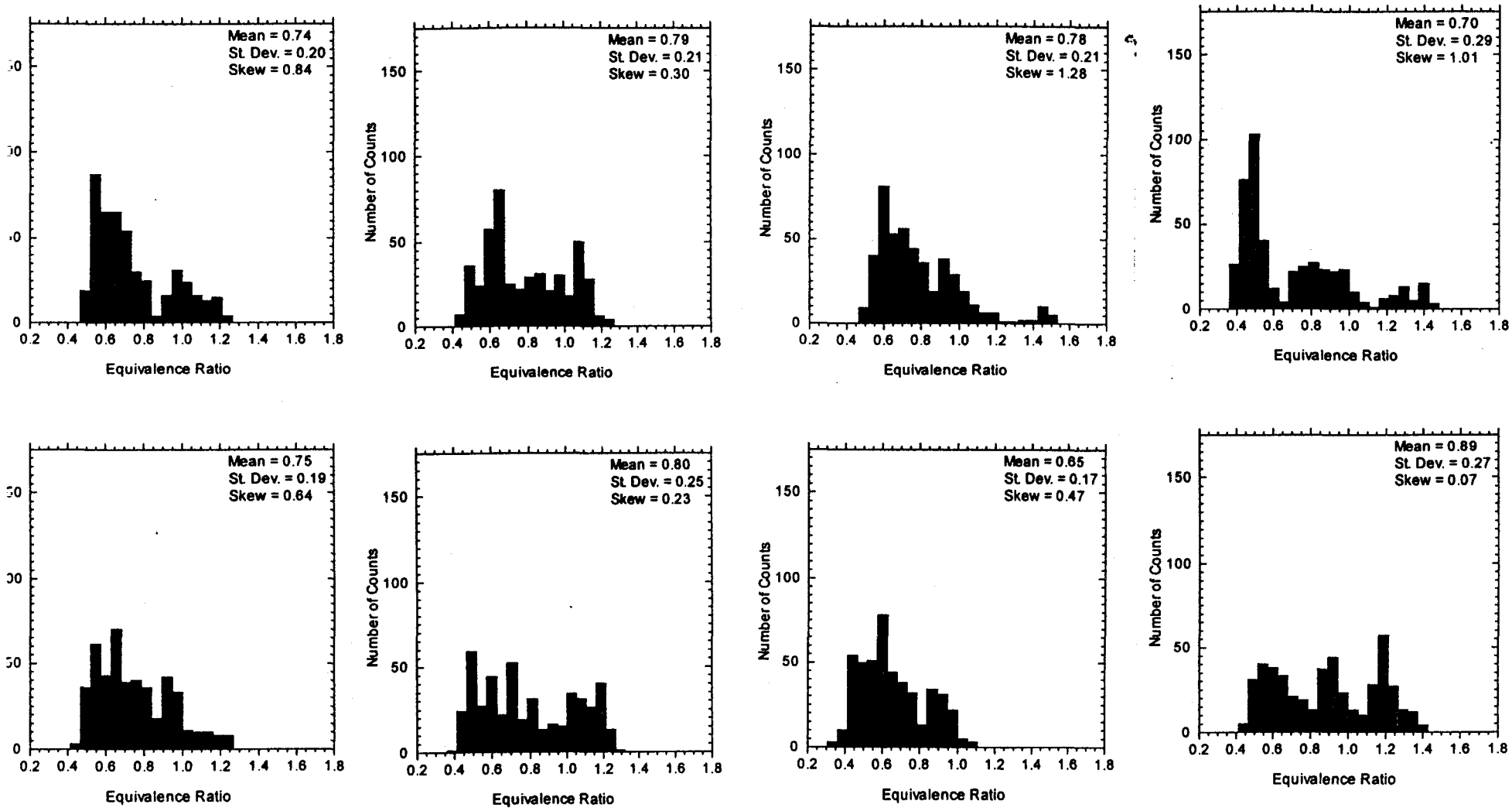


Figure 29 Measured local PDF's at selected points of a generic DLE mixer exit plane by Peters[70], inlet pressure = 16.7 psia, inlet temperature 430 K, 3.1% pressure drop, mean equivalence ratio = 0.7.